
Data Preprocessing

What is Data Preprocessing? — Major Tasks

☐ Data cleaning

- ☐ Handle missing data, smooth noisy data, identify or remove outliers, and resolve inconsistencies

☐ Data integration

- ☐ Integration of multiple databases, data cubes, or files

☐ Data reduction

- ☐ Dimensionality reduction
- ☐ Numerosity reduction
- ☐ Data compression

☐ Data transformation and data discretization

- ☐ Normalization
- ☐ Concept hierarchy generation

Why Preprocess the Data? — Data Quality Issues

- ❑ Measures for data quality: A multidimensional view
 - ❑ Accuracy: correct or wrong, accurate or not
 - ❑ Completeness: not recorded, unavailable, ...
 - ❑ Consistency: some modified but some not, dangling, ...
 - ❑ Timeliness: timely update?
 - ❑ Believability: how trustable the data are correct?
 - ❑ Interpretability: how easily the data can be understood?

Data Cleaning

- ❑ **Data in the Real World Is Dirty:** Lots of potentially incorrect data, e.g., instrument faulty, human or computer error, and transmission error
 - ❑ Incomplete: lacking attribute values, lacking certain attributes of interest, or containing only aggregate data
 - ❑ e.g., *Occupation* = " " (missing data)
 - ❑ Noisy: containing noise, errors, or outliers
 - ❑ e.g., *Salary* = "-10" (an error)
 - ❑ Inconsistent: containing discrepancies in codes or names, e.g.,
 - ❑ *Age* = "42", *Birthday* = "03/07/2010"
 - ❑ Was rating "1, 2, 3", now rating "A, B, C"
 - ❑ discrepancy between duplicate records
 - ❑ Intentional (e.g., *disguised missing data*)
 - ❑ Jan. 1 as everyone's birthday?

Data Cleaning

❑ Importance

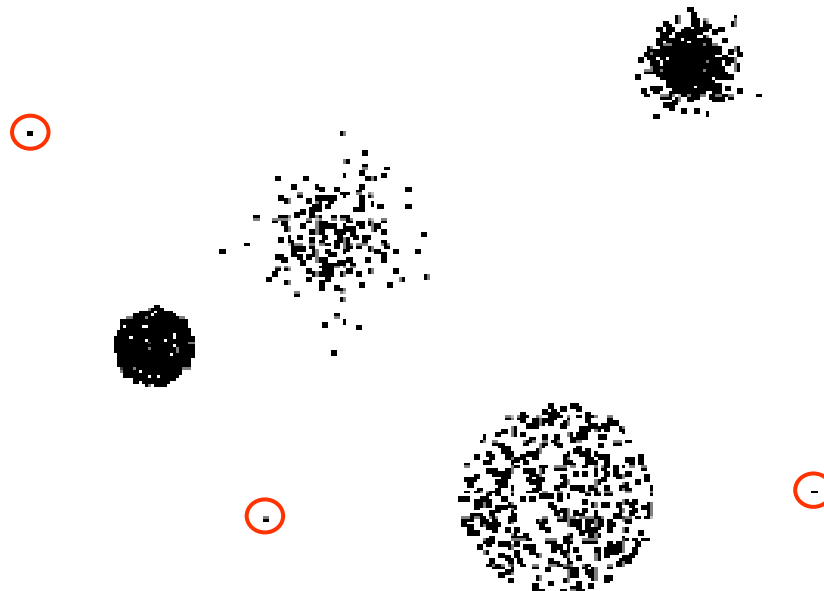
- ❑ “Data cleaning is one of the three biggest problems in data warehousing

❑ Data cleaning tasks

- ❑ Identify outliers and smooth out noisy data
- ❑ Fill in missing values
- ❑ Correct inconsistent data
- ❑ Resolve redundancy caused by data integration

1-Outliers

- ❑ Outliers are data objects with characteristics that are considerably different than most of the other data objects in the data set



2- Incomplete (Missing) Data

- ❑ Data is not always available
 - ❑ E.g., many tuples have no recorded value for several attributes, such as customer income in sales data
- ❑ Missing data may be due to
 - ❑ Equipment malfunction
 - ❑ Inconsistent with other recorded data and thus deleted
 - ❑ Data were not entered due to misunderstanding
 - ❑ Certain data may not be considered important at the time of entry
 - ❑ Did not register history or changes of the data
- ❑ Missing data may need to be inferred

How to Handle Missing Data?

- ❑ Ignore the tuple: usually done when class label is missing (when doing classification)—not effective when the % of missing values per attribute varies considerably
- ❑ Fill in the missing value manually: tedious + infeasible?
- ❑ Fill in it automatically with
 - ❑ a global constant : e.g., “unknown”, a new class?!
 - ❑ the attribute mean
 - ❑ the attribute mean for all samples belonging to the same class: smarter
 - ❑ **the most probable value: inference-based such as Bayesian formula or decision tree**

3-Duplicate Data (redundancy)

- ❑ Data set may include data objects that are duplicates, or almost duplicates of one another
 - ❑ Major issue when merging data from heterogenous sources
- ❑ Examples:
 - ❑ Same person with multiple email addresses

4- Noisy Data

- ❑ **Noise:** random error or variance in a measured variable
- ❑ **Incorrect attribute values** may be due to
 - ❑ Faulty data collection instruments
 - ❑ Data entry problems
 - ❑ Data transmission problems
 - ❑ Technology limitation
 - ❑ Inconsistency in naming convention
- ❑ **Other data problems**
 - ❑ Duplicate records
 - ❑ Incomplete data
 - ❑ Inconsistent data

How to Handle Noisy Data?

- ❑ Binning

- ❑ First sort data and partition into (equal-frequency) bins
 - ❑ Then one can **smooth by bin means, smooth by bin median, smooth by bin boundaries**, etc.

This is also called 'discretizing' a numeric attribute.

- ❑ Regression

- ❑ Smooth by fitting the data into regression functions

- ❑ Clustering

- ❑ Detect and remove outliers

- ❑ Semi-supervised: Combined computer and human inspection

There are two types of discretization: ***equal interval and equal frequency***.

- ❑ first sort data and partition into (equi-depth) bins then smooth by bin means, smooth by bin median, smooth by bin boundaries, etc.

- ❑ **Equal-width** (distance) partitioning:
 - ❑ Divides the range into N intervals of equal size: uniform grid
 - ❑ if A and B are the lowest and highest values of the attribute, the width of intervals will be:
$$W = (B - A) / N.$$
 - ❑ The most straightforward, but outliers may dominate presentation
 - ❑ Skewed data is not handled well.

- ❑ **Equal-depth** (frequency) partitioning:
 - ❑ Divides the range into N intervals, each containing approximately same number of samples
 - ❑ Good data scaling
 - ❑ Managing categorical attributes can be tricky.

Binning Methods for Data Smoothing

- ❑ Sorted data (e.g., by price)
 - ❑ 4, 8, 9, 15, 21, 21, 24, 25, 26, 28, 29, 34

- ❑ Partition into (equi-depth) bins:
 - ❑ Bin 1: 4, 8, 9, 15
 - ❑ Bin 2: 21, 21, 24, 25
 - ❑ Bin 3: 26, 28, 29, 34

- ❑ Smoothing by bin means:
 - ❑ Bin 1: 9, 9, 9, 9
 - ❑ Bin 2: 23, 23, 23, 23
 - ❑ Bin 3: 29, 29, 29, 29

- ❑ Smoothing by bin boundaries:
 - ❑ Bin 1: 4, 4, 4, 15
 - ❑ Bin 2: 21, 21, 25, 25
 - ❑ Bin 3: 26, 26, 26, 34

Data Preprocessing

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- ❑ Data integration
- ❑ Data transformation
- ❑ Data reduction

Discretization and concept hierarchy generation

Data Integration

- ❑ Data integration:

- ❑ combines data from multiple sources

- ❑ Schema integration

- ❑ integrate metadata from different sources

1- Entity identification problem:

- ❑ identify real world entities from multiple data sources,
e.g., $A.cust-id \equiv B.cust-\#$

- ❑ Detecting and resolving data value conflicts
- ❑ for the same real-world entity, attribute values from different sources are different, e.g., different scales, metric vs. British units
- ❑ possible reasons: different representations, different scales, e.g., metric vs. British units

2- Redundant data problem

- ❑ The same attribute may have different names in different databases
- ❑ One attribute may be a “derived” attribute in another table, e.g., annual revenue
- ❑ Redundant data may be able to be detected by **correlational analysis**
- ❑ Careful integration of the data from multiple sources may help reduce/avoid redundancies and inconsistencies and improve mining speed and quality

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Discretization and concept hierarchy generation

Data Transformation

- ☐ Normalization
- ☐ Discretization
- ☐ Data Compression
- ☐ Sampling

Data Transformation

- ❑ A function that maps the entire set of values of a given attribute to a new set of replacement values s.t. each old value can be identified with one of the new values
- ❑ Methods
 - ❑ Smoothing: Remove noise from data
 - ❑ Attribute/feature construction
 - ❑ New attributes constructed from the given ones
 - ❑ Aggregation: Summarization, data cube construction
 - ❑ Normalization: Scaled to fall within a smaller, specified range
 - ❑ min-max normalization
 - ❑ z-score normalization
 - ❑ normalization by decimal scaling
 - ❑ Discretization: Concept hierarchy climbing

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- ❑ **Data reduction**

Discretization and concept hierarchy generation

Data Reduction Strategies

- ❑ A data warehouse may store terabytes of data
 - ❑ Complex data analysis/mining may take a very long time to run on the complete data set
- ❑ Data reduction
 - ❑ Obtain a reduced representation of the data set that is much smaller in volume but yet produce the same (or almost the same) analytical results
- ❑ Data reduction strategies
 - ❑ **Dimensionality reduction** — remove unimportant attributes
 - ❑ **Data Compression**
 - ❑ **Numerosity reduction** — used to choose smaller forms of data representation for reducing the dataset volume
 - ❑ **Discretization** and concept hierarchy generation